



Fig. 2.7 In a “processed method” the inputs are first transformed before repeated application of a given method; after post-processing, the output may have higher order of accuracy than would be obtained using a straightforward application of the scheme

The use of conjugacy as a method to develop enhanced numerical methods was introduced in [397] under the name *symplectic correctors*.

2.5 Other Classes of Methods

2.5.1 Runge-Kutta Methods

The family of Runge-Kutta methods for solving $\dot{z} = f(z)$ is defined by

$$Z = z + h \sum_{i=1}^s b_i F_i,$$

where the vectors F_i , $i = 1, \dots, s$, are computed by solving the system

$$F_i = f(z + h \sum_{j=1}^s a_{ij} F_j), \quad i = 1, \dots, s.$$

The parameters s , $b_1, b_2, \dots, b_s$, and the $s \times s$ matrix of coefficients $A = (a_{ij})$ describe the method. In some cases the determination of the vectors F_i requires the solution of a nonlinear system, i.e. the method is *implicit*; in other cases the method can be explicit. An example of a popular 4th order explicit method is the choice of matrix A with coefficients $a_{ij} \equiv 0$ except $a_{21} = 1/2$, $a_{32} = 1/2$ and $a_{43} = 1$, and $b_1 = 1/6$, $b_2 = 1/3$, $b_3 = 1/3$, $b_4 = 1/6$. This method is not symplectic, and it is in fact impossible to find symplectic explicit methods within the Runge-Kutta family.

Let us emphasize that, while a typical RK method is not symplectic, **some** implicit Runge-Kutta methods **are** symplectic. The precise condition that must be

satisfied [325] is

$$b_i a_{ij} + b_j a_{ji} = b_i b_j, \quad i = 1, \dots, s, \quad j = 1, \dots, s.$$

The implicitness reduces the usefulness of Runge-Kutta methods for applications like molecular dynamics, because the complexity and cost of the molecular force laws usually means that implicit methods are inefficient (this is certainly not *prima facie* obvious, but the issue has been examined by several researchers without much positive result; see, e.g. [308]).

Example 2.7 The Gauss-Legendre family of Runge-Kutta (GLRK) methods correspond to approximating the vector field at the Gauss points, i.e. the zeros of the orthogonal polynomials that arise in Gaussian quadrature. As these points are symmetrically distributed the GLRK schemes are symmetric, hence have even order. The simplest such method is the *implicit midpoint* rule:

$$\mathbf{Z} = \mathbf{z} + h\mathbf{F}_1, \quad \mathbf{F}_1 = \mathbf{f}\left(\mathbf{z} + \frac{h}{2}\mathbf{F}_1\right),$$

which has order 2. The 4th order method ($s = 2$) has coefficients

$$b_1 = b_2 = \frac{1}{2}, \quad A = (a_{ij}) = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} - \frac{\sqrt{3}}{6} \\ \frac{1}{4} + \frac{\sqrt{3}}{6} & \frac{1}{4} \end{bmatrix}.$$

2.5.2 Partitioned Runge-Kutta Methods

It can be seen that Runge-Kutta methods treat all components of the differential equation identically. On the other hand, in molecular dynamics the equations of motion often have a special structure: for example the differential equation system is typically linear in $\mathbf{p}$, and, moreover, the equations have a special coupling structure so that the differential equation for $\mathbf{q}$ depends only on $\mathbf{p}$ and that for $\mathbf{p}$ depends only on $\mathbf{q}$. So-called partitioned Runge-Kutta methods allow us to exploit this structure. As an illustration, consider the method:

$$\hat{\mathbf{P}} = \mathbf{p} - \frac{h}{2} \nabla_{\mathbf{q}} H(\mathbf{q}, \hat{\mathbf{P}}), \tag{2.26}$$

$$\mathbf{Q} = \mathbf{q} + \frac{h}{2} (\nabla_{\mathbf{p}} H(\mathbf{q}, \hat{\mathbf{P}}) + \nabla_{\mathbf{p}} H(\mathbf{Q}, \hat{\mathbf{P}})), \tag{2.27}$$

$$\mathbf{P} = \hat{\mathbf{P}} - \frac{h}{2} \nabla_{\mathbf{q}} H(\mathbf{Q}, \hat{\mathbf{P}}). \tag{2.28}$$